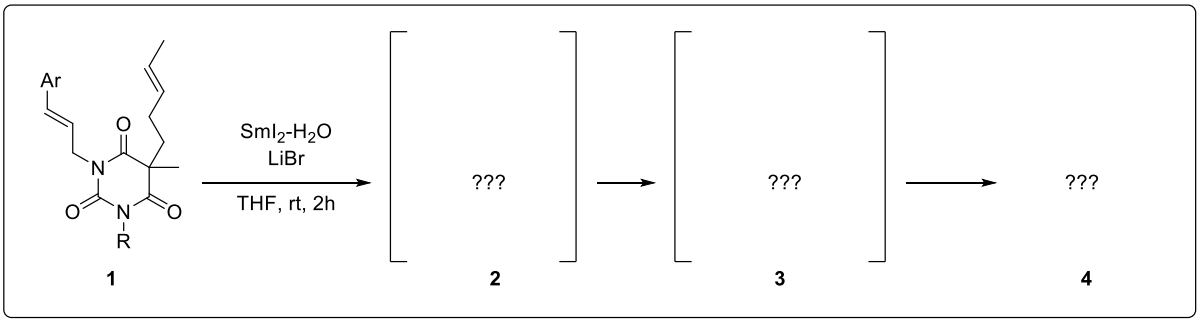
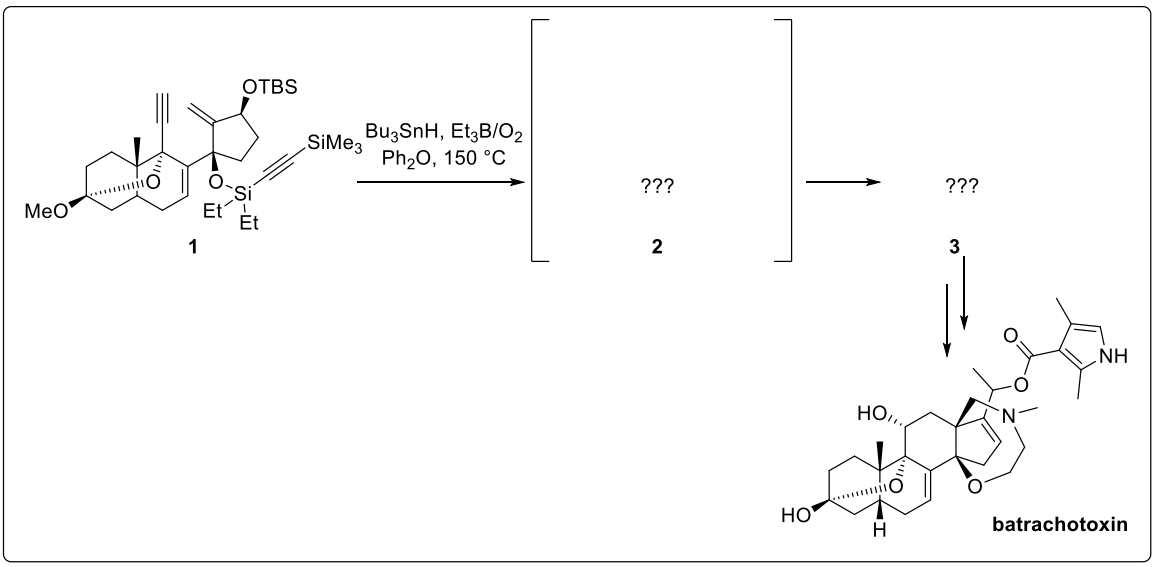
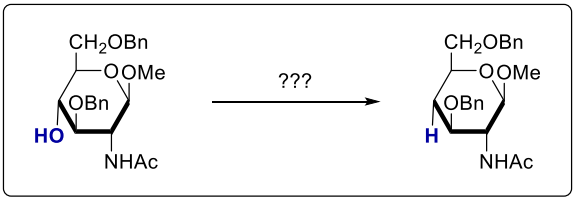
1. **Barbiturate-type substrate 1** was converted to a complex product via a radical cascade reaction. Draw the structures of the intermediates and final product and provide a reaction mechanism.



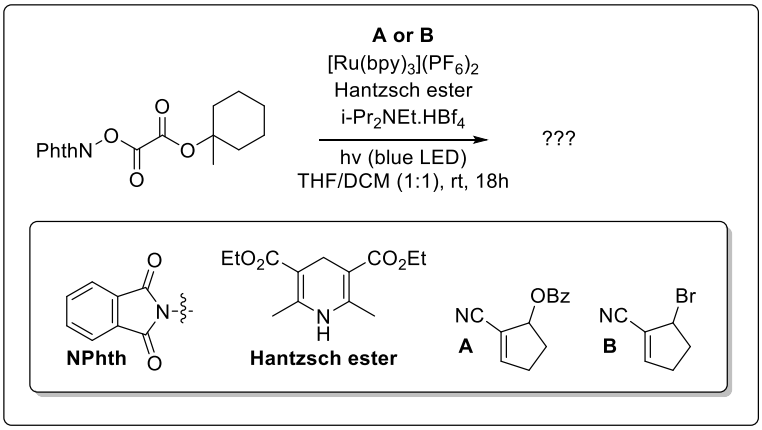
2. **Compound 1** is an advanced intermediate in the total synthesis of **batrachotoxin**. Given the reaction conditions propose a structure for the unstable **intermediate 2** and subsequent **intermediate 3**.



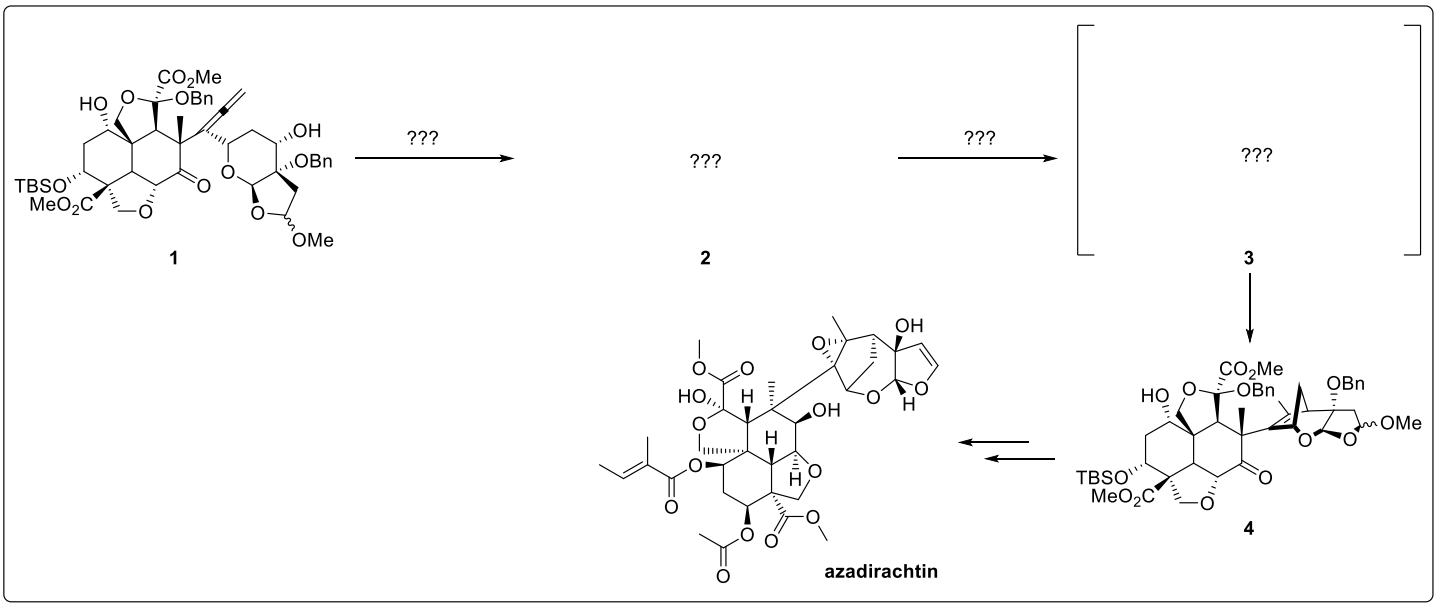
3. Propose a radical strategy for the replacement of the hydroxyl group by a hydrogen atom? Draw the mechanism of the radical step showing the initiation, propagation, and potential termination steps.



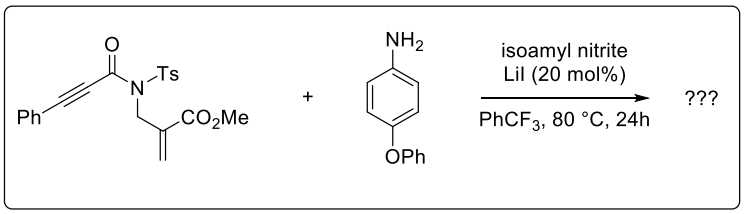
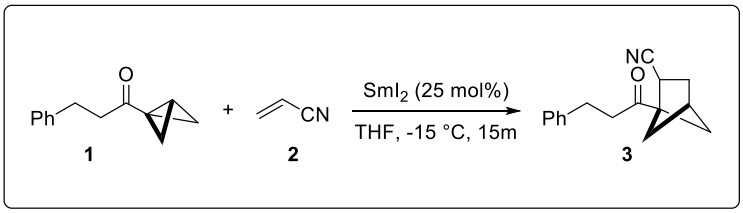
4. Propose a product and mechanism for both the reaction with **compound A** and **B** under otherwise identical conditions.



5. **Compound 1** is an advanced intermediate in the total synthesis of **azadirachtin**. Propose a radical strategy to reach the desired **intermediate 4**. Show the structures of the intermediates and propose reaction conditions.



6. Propose a product and mechanism for the transformation shown below.



7. Propose a catalytic radical cascade mechanism for the transformation shown below.